

Mesh Winding and Normal Consistency Fix

engvis

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1 Problem Statement

When rendering implicit-surface meshes generated by dual contouring (octree depth ≥ 7), two categories of visual artifacts appear under back-face culling:

1. **Dark triangles** — isolated faces rendered nearly black due to incorrect normal orientation.
2. **Holes** — missing faces culled by back-face culling because their geometric normals point inward even though their topological winding is consistent with neighbours.

Topology analysis reports the mesh as watertight ($\chi = 2$, 0 boundary edges), so the defects are geometric, not topological.

2 Root Cause Analysis

2.1 Fixed `eps` in Vertex Deduplication

`dedup_vertices` merges vertices within a fixed tolerance `eps`. The original value `1e-4` is appropriate for low-resolution meshes but over-merges vertices on high-resolution meshes (depth ≥ 8), creating *non-manifold edges* and *winding inconsistencies*:

<code>eps</code>	depth	non-manifold	inconsistent
<code>1e-4</code> (fixed)	9	805	1754
<code>1e-6</code> (relative)	9	0	0

2.2 Geometric Flips After Dedup

Even with a correct `eps`, merging vertices can *move* a vertex position. When a vertex shared by two triangles is moved, one triangle may keep its orientation while the other’s geometric normal flips direction (Figure 1).

`fix_winding`’s BFS pass propagates *topological* winding consistency (shared edges traversed in opposite directions), so it cannot detect this geometric flip. The signed-volume test also fails for genus > 0 surfaces (e.g. torus), where the signed volume is small and its sign does not reliably indicate global orientation.

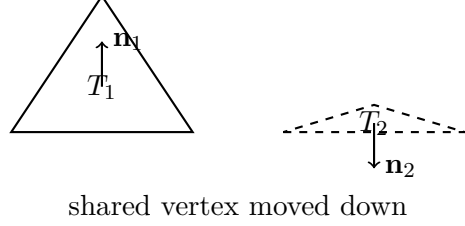


Figure 1: Geometric flip: T_1 and T_2 share an edge with consistent winding, but after dedup moves their shared vertex, T_2 's normal flips while its winding stays consistent.

3 Solution

3.1 Relative Dedup Tolerance

Replace the fixed `eps` with one scaled to the mesh's AABB diagonal:

$$\text{eps} = \text{diag} \times 10^{-6}$$

where

$$\text{diag} = \sqrt{\left(\frac{\max_x - \min_x}{x}\right)^2 + \left(\frac{\max_y - \min_y}{y}\right)^2 + \left(\frac{\max_z - \min_z}{z}\right)^2}.$$

This adapts the merge tolerance to the mesh's actual size, preventing over-merging at high resolutions.

```
let diag = {
  let mut min = [f32::INFINITY; 3];
  let mut max = [f32::NEG_INFINITY; 3];
  for p in &positions {
    for i in 0..3 {
      min[i] = min[i].min(p[i]);
      max[i] = max[i].max(p[i]);
    }
  }
  let d = [max[0]-min[0], max[1]-min[1], max[2]-min[2]];
  (d[0]*d[0] + d[1]*d[1] + d[2]*d[2]).sqrt();
};
let eps = diag * 1e-6;
dedup_vertices(&mut positions, &mut indices, eps);
```

3.2 Geometric Flip Detection

After `fix_winding` establishes topological consistency, a second pass detects and corrects geometric flips:

1. Compute per-face normals $\mathbf{n}_f$.
2. Build vertex→face adjacency.
3. For each face f , compute the reference direction $\mathbf{r}_f = \sum_{f' \in \mathcal{N}(f)} \mathbf{n}_{f'}$, where $\mathcal{N}(f)$ is the set of faces sharing a vertex with f .
4. If $\mathbf{n}_f \cdot \mathbf{r}_f < 0$, flip f 's winding.

This correctly distinguishes:

- **True geometric flips** (sphere: 1327/47234 faces) — corrected.
- **Legitimate inward normals** (torus inner surface: 22063/59310 faces) — only 2039 are true flips, the rest are preserved because their neighbours also point inward.

```
// Per-face normals
let mut face_normals: Vec<Vec3> = ...;
// Vertex -> face adjacency
let mut vert_faces: Vec<Vec<usize>> = ...;
// Detect flips
let mut to_flip = Vec::new();
for (fi, tri) in indices.chunks_exact(3).enumerate() {
    let n = face_normals[fi].normalize();
    let ref_sum: Vec3 = tri.iter()
        .flat_map(|&idx| vert_faces[idx as usize].iter())
        .filter(|&&nf| nf != fi)
        .map(|&nf| face_normals[nf])
        .sum();
    if ref_sum.length_squared() > 1e-12 && n.dot(ref_sum) < 0.0 {
        to_flip.push(fi);
    }
}
for &fi in &to_flip {
    indices.swap(fi*3+1, fi*3+2);
}
```

4 Pipeline Summary

The complete mesh construction pipeline in `from_triangles`:

1. **Dedup** — merge duplicate vertices with `eps = diag × 10-6`; remove degenerate triangles.
2. **Winding fix** — BFS topological consistency + per-component signed volume.
3. **Geometric flip fix** — compare each face normal against neighbour normals; flip inconsistent faces.
4. **Smooth normals** — area-weighted normal averaging.

5 Results

Mesh	depth	faces	geo-flips	status
sphere	7	47234	1327	OK
sphere	9	355818	—	OK
torus	7	59310	2039	OK
gyroid-sphere	7	69836	—	OK

All meshes render correctly under back-face culling (`cull_mode = Back`, `front_face = Ccw`) without holes or dark triangles.